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Systematic approach to understanding the pathogenesis of systemic sclerosis

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Conflict of interest statement

The authors declare no competing financial interests.

Author Contributions statement

Dr. XZ and Dr. LZ participated in the study design and drafted the manuscript. Dr. HL participated in the English language editing. Dr. YL and Dr. HZ conceived the study, revised and finalized the manuscript. All authors read and approved the final manuscript.

Acknowledgements

This work was supported by grants from the National Natural Science Foundation of China (81373206, 81401357), National High Technology Research and Development Program of China (2012AA02A513), Hunan Provincial Natural Science Foundation of China (13JJ5010), and Hunan Special Fund for Science and Technology Development Program (2013WK3022).

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/cge.12946

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Abstract

Systemic sclerosis (SSc) is a complex heterogeneous autoimmune disease. Progressive organ fibrosis is a major contributor to SSc mortality. Despite extensive efforts, the underlying mechanism of SSc remains unclear. Efforts to understand the pathogenesis of SSc have included genomics, epigenetics, transcriptomic, proteomic and metabolomic studies in the last decade. This review focuses on recent studies in SSc research based on multi-omics. The combination of these technologies can help us understand the pathogenesis of SSc. This review aims to provide important information for disease identification, therapeutic targets and potential biomarkers.

Keywords: Systemic sclerosis; pathogenesis; multi-omics; Combination

Introduction

Systemic sclerosis (SSc) is a complex heterogeneous fibrosing autoimmune disease with an unknown pathogenesis. Clinically, SSc can be classified into 2 major subtypes: limited SSc (lSSc), defined as fibrosis distal to the elbow and the knee joint (including the face), and the diffuse form (dSSc) with proximal fibrosis(1). LSSc is more commonly associated with pulmonary arterial hypertension (PAH), whereas dSSc is more prominently linked to interstitial lung diseases (ILD)(2). PAH and ILD are the two most serious complications and major causes of death in SSc patients.

The exact cause of SSc has not been completely elucidated but likely involves environmental factors in a genetically primed individual(3). The fibroblast activation and fibrosis underlying SSc are induced by vascular injury and endothelial activation, leading to an uncontrolled inflammatory reaction and subsequent irreversible scarring and organ failure. The molecular mechanisms of fibroblast activation in SSc are complicated. Transforming growth factor-β signalling has a central role by inducing profibrotic gene expression through activation of canonical (SMAD2–SMAD3 (SMAD2/3)- and SMAD4‑dependent) and non-canonical pathways and through crosstalk with other intracellular signalling pathways. In addition,

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canonical WNT signalling, the Toll-like receptor 4 signalling pathway, mechanical stress, hypoxia, reactive oxygen species generation, damage-associated molecular patterns, connective tissue growth factor, tenascin-C, platelet-derived growth factor, IL-6 and IL-13, among other factors, drive collagen production and promote fibrotic matrix deposition.

Omics analysis is a powerful way to analyse disease processes at the cellular level(4), and with the development of high-throughput technologies, integrative systems biology approaches have greatly improved our understanding of the underlying mechanisms associated with disorders(5, 6). This technology has been widely used to study many tissues and cells of SSc. However, most studies have focused on a single omics method, and very few studies have performed integrated multi-omics analysis to clarify the pathogenesis of SSc. A combination of genetic association studies, epigenetics, gene expression profiles, proteomics and metabolomics may facilitate the elucidation of the pathophysiology of SSc and the development of valuable biomarkers.

Genetics of SSc

Numerous genome-wide association studies (GWAS) have been conducted, and many susceptibility regions have been identified in SSc. The HLA locus remains the only common and confirmed susceptibility locus for SSc, but other non-HLA susceptibility genes have also been identified recently.

GWAS identified single nucleotide polymorphisms (SNPs) within many genes as SSc risk loci, such as the HLA-DPB1 and DPB2 genes in the Korean and US Caucasian populations (7), interferon regulatory factor (IRF) 5, STAT4 and 5 non-HLA loci, CD247 in populations of Caucasian and European ancestry (8), PSORS1C1, TNIP1 and RHOB loci in Europe (9, 10), DNASE1L3, SCHIP1-IL12A, ATG5 and TREH-DDX6 loci in a mixed population (US, Canada, Spain, Germany, the Netherlands, Italy, Sweden, and the UK)(11), LRRC34 and OBFC1 in a Caucasian population (12), and ATP8B4, ASB10, CNGB3, HLA-DRB5 and HSPB2 in European-American subjects (13). Replication studies further validated that CD247 and TNIP1 were correlated with SSc susceptibility but not PSORS1C1 and RHOB (8, 10). Pan-meta-analysis and GWAS follow-up studies identified KIAA0319, PXK and JAZF1 in the UK and Spanish population (14), IL12RB2 in the US population (15), PPARG in European ancestry

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(Italy and the United Kingdom) (16), CSK in European ancestry (17) and IRF4 in European ancestry(18) as risk loci for SSc.

SNPs are associated not only with SSc susceptibility but also with clinical phenotypes. Positive associations were found between the HLA-DPB1and DPB2 loci and anti-DNA topoisomerase I autoantibodies (ATA) and anti-centromere autoantibodies (ACA) in the Korean and US Caucasian populations(7), IRF8 and GRB10 and lSSc, SOX5 and HLA-DQB1 and ACA, HLA-DPA1/B1 and ATA, NOTCH4 and ACA and ATA in the Caucasian population(19), MUC2 and SSc-ILD, DSP, SPPL2C and MAPT and the percent of predicted forced vital capacity (FVC%) in the Caucasian population (12), and ATP8B4 and PAH in European-American subjects (13). Taken together, these findings indicate that genetic factors play an essential role in SSc aetiology. These SNPs identified by GWAS can be divided into HLA and non-HLA genes. HLA class II molecules (HLA-DP, HLA-DQ, and HLA-DR) vary among different populations and are related to serum autoantibodies. The non-HLA genes are involved in the Type I IFN pathway, IL-12 pathway, T cell signalling pathway, TNF signalling pathway, and the clearance, autophagy, and detoxification of debris(20).

Global epigenetics and SSc

Although genetic factors play important roles in the development of SSc, the concordance rate of SSc in monozygotic twins is only 4.2%(21). Growing evidence suggests gene-specific epigenetic alterations in SSc, thus indicating a possible role of environmental–epigenetic factors in the pathogenesis of SSc(22). Epigenetic mechanisms include DNA methylation, histone modification, miRNA and long non-coding RNA. Here, we provide a comprehensive overview of microarray-based epigenetic modifications in SSc patients.

Global methylation analysis has been performed in CD4+ T cells, peripheral lymphocytes and dermal fibroblasts of SSc. In CD4+ T cells, DNA methylation enzymes such as DNMT1, MBD3 and MBD4 are decreased in SSc(23). In peripheral lymphocytes from monozygotic twins discordant for SSc, the methylation status of X chromosome genes including transcription factors (ARX, HSFX1, ZBED1 and ZNF41), surface antigens (IL1RAPL2 and PGRMC1), and genes involved in cell proliferation (PGK1, SMS, UTP14A and SSR4), apoptosis (MTM1),

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inflammation (ARAF) and oxidative stress (ENOX2) was associated with SSc (24). In dermal fibroblasts, dSSc and lSSc showed distinct and characteristic methylation patterns. A significant correlation between DNA methylation status and gene expression was found. Differentially methylated genes involved in extracellular matrix–receptor interactions and focal adhesion and many fibrosis-related genes (ITGA9, ADAM12, COL23A1, COL4A2, MYO1E, RUNX1, RUNX2 and RUNX3) are hypomethylated in both dSSc and lSSc(25).

Histone modifications include the acetylation, methylation, and phosphorylation of specific histone residues and influence the accessibility of chromatin to transcription factors at gene promoters and enhancers. Histone acetylation is tightly controlled by the balance of two antagonizing histone-modifying enzymes, histone acetyltransferase (HAT) and deacetylase (HDAC)(26). Aberrant histone modifications have been correlated with skin thickness and SSc activity. Global histone H4 is hyperacetylated and H3K9 is hypomethylated in SSc peripheral blood B cells, affecting the development and activation of B cells and playing important roles in the pathogenesis of SSc(27).

Many miRNAs are involved in the autoimmune, vascular and fibrotic processes of SSc(28). In SSc lesion skin, characteristic miRNA patterns have been identified, and most of the differentially expressed miRNAs were correlated with SSc fibrosis (28, 29). In SSc fibroblasts and TGF-β-stimulated normal fibroblasts, 56 dysregulated miRNAs were identified in common (49 upregulated and 7 downregulated miRNAs). Among them, miR-196a inhibits the expression of type I collagen. Lower serum miR-196a levels were associated with a higher ratio of dSSc:lSSc, higher modified Rodnan total skin thickness score, and higher prevalence of pitting scars(30). Serum miR-206 and miR-21 helped distinguish SSc patients from normal subjects(31). MiRNA-223, -181b, -342-3p and -184 were differentially expressed in the lSSc and dSSc groups, and miRNA-409, -184, -92a, -29a and -101 exhibited greatly different levels in one or more autoantibody groups. Specific miRNAs were associated with autoantibody profiles of known diagnostic and prognostic value (32). A 21-miRNA combined model was constructed with optimum accuracy (80%) to differentiate SSc from both normal subjects and those with lupus. Serum miR-16, -223, and -638 differentiated SSc from normal controls; miR-142-3p, -150, -223,

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and -638 differentiated SSc from systemic lupus erythymatosus(SLE). MiR-638 showed a weak correlation with anti-Scl-70 autoantibody (33).

Transcriptomic profiles in SSc

SSc peripheral blood cells

Five studies have investigated global gene expression in SSc peripheral blood cells, including peripheral blood cells (PBCs), peripheral blood mononuclear cells (PBMCs), CD4+ T cells and monocytes (34-38). All of these studies revealed different gene expression profiles in SSc. The most important discovery was the IFN-inducible gene signature, which was found in all studies. MX1 was also found in all studies, whereas OAS1, OAS2, OAS3, IFIT1, IFIT2, IFIT3, IRF7, IFIH1, IFI44L, IFI30, PLSCR1, STAT1, TNFSF13B, RSAD2, SERPING1, and IFITM3 were found in more than 3 studies. The expression level of IFN-induced genes reflected the SSc clinical heterogeneity. Low expression of IFN response genes was associated with the presence of ACA, whereas increased expression was associated with the presence of anti-topoisomerase and anti-U1 RNP antibodies, lymphopenia and digital ulcers (36, 37).

SSc skin biopsies and skin fibroblasts

Distinct gene profiles in SSc skin biopsies were identified compared with normal controls, but no obvious expression difference was observed between clinically affected and unaffected skin. Most differentially expressed genes were involved in the Wnt and transforming growth factor beta (TGF-β) pathways (39-41). Based on the skin gene expression profiles, SSc can be divided into 4 major subgroups (proliferation, inflammation, limited, and normal-like subgroups) or 3 major subgroups (keratin, fibro-inflammatory, and normal-like subgroups). Each subgroup has a unique gene expression signature and correlates with clinical variables. For example, the proliferation subgroup is highly associated with the modified Rodnan skin score (mRSS), whereas the limited subgroup is highly associated with the severity of Raynaud's phenomenon. Higher keratin transcript scores were associated with ILD and shorter disease duration, higher fibro-inflammatory scores with higher mRSS, and a normal-like transcript pattern with longer disease duration(41, 42). These subgroups were intrinsic, inherent, stable and independent of

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disease duration and thus are helpful in guiding SSc treatment. In the rituximab treatment group, consistent and non-progressing gene expression was observed regardless of treatment regimen or the time point of measurement, consistent with an apparent lack of clinical responses(43). In the mycophenolate mofetil (MMF) treatment group, patients in the inflammatory signature subgroup showed clinical improvement during MMF treatment, but the fibroproliferative and normal-like subgroups showed no clinical improvement. The gene expression profile changed during MMF treatment only in the subjects who showed improvement. Thus, targeting inflammation is likely to benefit patients with an inflammatory signature, and targeting fibrosis may benefit patients with a fibroproliferative signature (44). Two models of multigene and pharmacodynamic biomarkers based on skin gene expression were also developed to guide treatment(45).

In skin fibroblasts, the number of differential genes was far smaller than that in skin biopsies (39, 46-49). There was no difference in gene expression profiles either between SSc lesion skin and non-lesion skin fibroblasts or between affected and unaffected monozygotic twins. Normal fibroblasts incubated with serum from an SSc patient or with serum from unaffected monozygotic twins twin showed increased expressions of COL1A2, SPARC, and CTGF, typically only in SSc fibroblasts(50). Explanted SSc fibroblasts incompletely reflected the SSc phenotypes, indicating that either nonfibroblast cell types or other aspects of the dermal milieu are required for the full expression of SSc phenotypes (50).

Lung tissues and fibroblasts form SSc ILD and PAH

ILD is the most serious complication and major cause of death in SSc patients. Global gene expression profiles have been performed in SSc lung tissues, fibroblasts and bronchoalveolar lavage fluid (BALF) cells. Similar to skin biopsies, SSc ILD tissues show inflammatory and fibrotic gene expression patterns. The differentially expressed genes are involved in the TGF-β and IFN signalling pathways and participate in pulmonary fibrosis, macrophage emigration and activation. Thus, blocking one or more of these pathways may be a good approach to treat SSc pulmonary complications (51, 52). In the gene expression profiles of SSc ILD lung fibroblasts, a TGF-β response signature and myofibroblast markers were upregulated with marked heterogeneity among the samples, whereas interferon-stimulated genes or interferon-related key

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regulators and mediators were decreased(53). In BALF cells, neutrophils, eosinophils, and lymphocytes were found. Chemokines and chemokine receptor genes were increased in the BALF cells of SSc with lung inflammation, leading to the migration of T cells and phagocytic cells and increasing the risk of lung fibrosis (54). CD8+ cells from the BALF cells produced profibrotic factors such as interleukin-4 and oncostatin M and directly promoted fibrosis (55). The inaccessibility of lung tissue precludes its widespread use in research and clinical practice, and thus recent studies have attempted to define predictive genes from skin. The correlation between skin gene expression and ILD severity has been studied (56). A total of 82 skin transcripts were correlated with FVC and distinguished patients with more severe ILD. These genes included SELP, CCL2, and matrix metalloproteinase 3, which are involved in the extravasation and adhesion of inflammatory cells. Eight of the FVC-correlated genes (CCL2, HAPLN3, GPR4, ADCYAP1, WARS, CDC25B, PLP1, and STXBP6) were correlated with mRSS.

In SSc PAH PBMCs, IFN-regulated genes, biomarker genes, inflammatory and immune-related genes were found as distinctive clusters. A combination of the gene expression of NOG, LRRN3, TGFβR2, IL-7R and CCR7 helped distinguish SSc PAH from SSc non-PAH. Several endoplasmic reticulum (ER) stress and unfolded protein response genes were extremely increased in lSSc PAH. MRC1 was increased exclusively in lSSc PAH and strongly correlated with pulmonary artery pressure and higher mortality (57-59). In lSSc PBMCs, the HSP gene DNAJB1 was positively correlated with the severity of PAH (PAP), and ER stress markers were correlated with IL-6 levels (60).

SSc oesophageal pathology

Oesophageal involvement is common in SSc patients, but the tissue-specific pathological mechanisms are poorly understood. Oesophageal biopsies have revealed unique gene expression profiles as well as inflammatory and proliferative gene expression signatures. The molecular classification of oesophageal biopsies was independent of SSc skin subtype, serum autoantibodies and esophagitis. The inflammatory signature was present in biopsies without a high count of infiltrating lymphocytes, which suggests that immune activation is a major driver

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of SSc oesophageal pathogenesis(61). All gene expression data are available in supplement table S1.

Proteomics in SSc

In SSc skin tissues, proteomic analysis revealed an altered abundance of proteins involved in extracellular matrix production, myofibroblast contractility, energy metabolism and response to oxidative stress. Proteins specific to the epidermis and involved in epidermal cell differentiation, such as keratins 14, 1/10, 6 and 16 were altered, suggesting SSc epidermis has an activated, wound-healing phenotype(62). In the serum of SSc, proteomic analysis identified 25 differentially expressed mass peaks, including S100A8, which was highest in SSc ILD patients (63). However, recombinant antibody microarrays found no obvious difference in the serum proteins of SSc compared with those in healthy controls, and only a few serological markers were differentially expressed, such as mucin-1 and MCP-4 (64). CXCL4 was secreted by plasmacytoid dendritic cells and upregulated in the circulation and skin. CXCL4 induced the influx of inflammatory cells and skin transcriptome changes in SSc, which correlated with the presence and progression of complications, such as ILD and PAH (65). In the BALF of SSc patients with lung fibrosis, glutathione S-transferase P (GSTP), Cu–Zn superoxide dismutase (SOD) and cystatin SN were downregulated, suggesting involvement in the mechanisms that protect the lungs from injury or inflammation. Haptoglobin-α chain, α1-acid glycoprotein, calgranulin (Cal) B, cytohesin-2, calumenin and mtDNA TOP1 were upregulated, indicating involvement in the induction of lung fibrogenesis(66). Upregulation of α2-macroglobulin, α1-antitrypsin and pulmonary surfactant protein A, and downregulation of α2 heat shock protein (HSP) and glutathione S-transferase (GST) were also identified in the BALF of SSc patients with lung fibrosis (67). In the secretome of cultured SSc dermal fibroblasts, 890 protein spots were differentially altered. These proteins represented a specific “fibrosis signature” reflecting an activated fibroblast phenotype with potential application as biomarkers for fibrotic diseases. Three reticulocalbin family members were the most up-regulated proteins. These calcium-binding proteins reside in the endoplasmic reticulum and participate in the trafficking and transport of newly synthesized proteins(68).

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Metabolome and SSc

The metabolome is the collection of small molecules (metabolites). Systematic identification and quantification of metabolites have been used to identify biomarkers for several diseases (69, 70). A total of 25 metabolites were identified as altered in SSc, and 3 metabolites (nonanoic acid, ribose and taurine) were specifically dysregulated in SSc compared with SLE and primary Sjögren's syndrome. These metabolite molecules could be used to identify biomarkers for SSc(71).

Discussion

SSc is a complex disease with pathophysiological changes in various organ systems. Over the past decade, advances in omics technologies and systems biology have greatly improved our understanding of the underlying mechanisms of SSc. Many omics-based approaches have been used in the diagnosis, evaluation, and prediction of treatment responses of SSc in clinical settings. However, most studies have used only a single omics approach and thus failed to fully discover the complex mechanisms of SSc. Integrated multi-omics data may facilitate the elucidation of functional principles at different levels of SSc, such as genes that are regulated at the chromatin, transcriptional, post-transcriptional, translational, and post-translational levels in the coordinated progression of the disease process. Furthermore, integration based on multi-omics data can not only provide insights into the physiological state but also identify new biomarkers for the diagnosis and the prognosis(72).

No study or review has completely analysed the pathogenesis of SSc using a multi-omics approach. This review is the first to combine all omics studies in SSc from genomics to metabolomics and may serve as a reference for future studies to identify the key underlying genetic and metabolic pathways.

Multi-omics may become a cornerstone enabling the exploration of the contribution of genotype and environment to phenotypic variations in SSc. Due to the diversity of SSc, it is important to

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expand the use of omics and systems biology methods to facilitate a more precise understanding of the mechanistic principles underlying SSc and improve SSc treatment in the future(73-75).

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Figure legends

Table 1 Selected SNPs identified as risk factors for SSc

Table 2 Important proteins identified in SSc

Table S1 Gene expression in SSc

Table S2 Abbreviation list for genes

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Table 1 Selected SNPs identified as risk factors for SSc

Gene	Clinical association	Population	Study approach	Refs
Major histocompatibility region genes (HLA-DPB1,DPB2)	SSc risk loci	Korean and US Caucasian population	GWAS	7
IRF5, STAT4, 5 non-HLA loci (CD247)	SSc risk loci	Caucasian and European ancestry	GWAS	8
PSORS1C1, TNIP1, RHOB	SSc risk loci	France, Italy, Germany, and Northern Europe；USA, Europe	GWAS and replicate study	9,10
DNASE1L3, SCHIP1-IL12A, ATG5,TREH-DDX6	SSc risk loci	US, Canada, Spain, Germany, the Netherlands, Italy, Sweden, and the UK	Immunochip Analysis	11
LRRC34,OBFC1	SSc risk loci	US	GWAS and replicate study	12
ATP8B4, ASB10, CNGB3, HLA-DRB5,HSPB2	SSc risk loci	European American	Whole-Exome Sequencing	13
KIAA0319, PXK,JAZF1	SSc risk loci	UK and Spain population	Pan-meta-analysis of GWAS study	14
IL12RB2	SSc risk loci	Caucasian populations	GWAS follow-up study	15
PPARG	SSc risk loci	European ancestry (Italy and United Kingdom)	GWAS follow-up study	16
CSK	SSc risk loci	European ancestry	GWAS follow-up study	17
IRF4	SSc risk loci	European ancestry	Meta-analysis of GWAS study	18
HLA-DPB1, DPB2	Positively with ATA and ACA	Korean and US Caucasian population	GWAS	7
IRF8, GRB10	lSSc	Caucasian population	GWAS meta-analysis stiudy	19
SOX5,HLA-DQB1,NOTCH4	Positively with ACA	Caucasian population	GWAS meta-analysis stiudy	19
HLA-DPA1/B1, NOTCH4	Positively with ATA	Caucasian population	GWAS meta-analysis stiudy	19
MUC2	SSc-ILD	US	GWAS and replicate study	12
DSP, SPPL2C, MAPT	Positively with FVC%	US	GWAS and replicate study	12
ATP8B4	PAH	European American	Whole-Exome Sequencing	13

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Table 2 Important proteins identified in SSc

Protein	Clinical association	Location	Study approach	Refs
keratin 14, keratin 1/10, keratins 6, keratin 16 ↑	epidermal cell differentiation	SSc skin tissues	proteomic analysis	62
S100A8↑	SSc ILD	serum	proteomic analysis	63
mucin-1 , MCP-4 ↑	SSc	serum	recombinant antibody microarrays	64
CXCL4↑	ILD and PAH	plasmacytoid dendritic cells, serum, skin	proteomic analysis	65
GSTP,SOD and cystatin SN ↓, Haptoglobin-α chain,α1-acid glycoprotein, calgranulinB, cytohesin-2, calumenin, mtDNA TOP1 ↑	inflammation and lung fibrogenesis	BALF	proteomic analysis	66
α2-macroglobulin, α1-antitrypsin and pulmonary surfactant protein A ↑, α2 -HSP, GST↓	lung fibrosis	BALF	proteomic analysis	67
Three reticulocalbin family members ↑	trafficking and transporting of newly synthesized proteins	SSc dermal fibroblasts	Proteomic Analysis	68

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